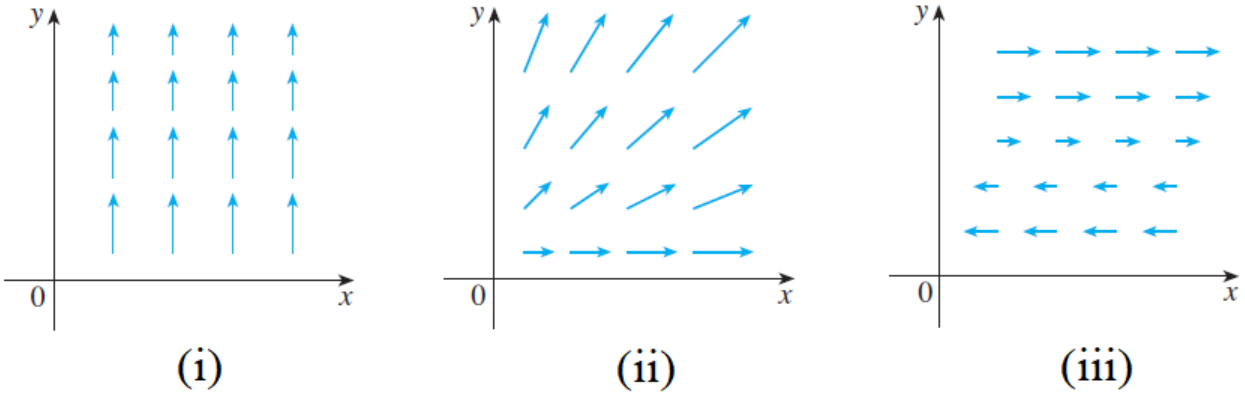
Vector Calculus and Complex Analysis Homework 2

1. Consider the following vector fields $\boldsymbol{F}$ in the xy-plane as shown in figure (i), (ii) and (iii).



(a) Is div($\boldsymbol{F}$) positive, negative, or zero? Explain.

(b) Determine whether curl($\boldsymbol{F}$) $= 0$. If not, in which direction does curl($\boldsymbol{F}$) point?

Solution

1(i) If the vector field is $\mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j} + R\,\mathbf{k}$, then we know $R = 0$. In addition, the x-component of each vector of $\mathbf{F}$ is 0, so $P = 0$, hence $\dfrac{\partial P}{\partial x} = \dfrac{\partial P}{\partial y} = \dfrac{\partial P}{\partial z} = \dfrac{\partial R}{\partial x} = \dfrac{\partial R}{\partial y} = \dfrac{\partial R}{\partial z} = 0$. Q decreases as y increases, so $\dfrac{\partial Q}{\partial y} < 0$, but Q doesn't change in the x- or z-directions, so $\dfrac{\partial Q}{\partial x} = \dfrac{\partial Q}{\partial z} = 0$.

(a) $\operatorname{div}\mathbf{F} = \dfrac{\partial P}{\partial x} + \dfrac{\partial Q}{\partial y} + \dfrac{\partial R}{\partial z} = 0 + \dfrac{\partial Q}{\partial y} + 0 < 0$

(b) $\operatorname{curl}\mathbf{F} = \left(\dfrac{\partial R}{\partial y} - \dfrac{\partial Q}{\partial z}\right)\mathbf{i} + \left(\dfrac{\partial P}{\partial z} - \dfrac{\partial R}{\partial x}\right)\mathbf{j} + \left(\dfrac{\partial Q}{\partial x} - \dfrac{\partial P}{\partial y}\right)\mathbf{k} = (0-0)\,\mathbf{i} + (0-0)\,\mathbf{j} + (0-0)\,\mathbf{k} = \mathbf{0}$

(ii) If the vector field is $\mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j} + R\,\mathbf{k}$, then we know $R = 0$. In addition, P and Q don't vary in the z-direction, so $\dfrac{\partial R}{\partial x} = \dfrac{\partial R}{\partial y} = \dfrac{\partial R}{\partial z} = \dfrac{\partial P}{\partial z} = \dfrac{\partial Q}{\partial z} = 0$. As x increases, the x-component of each vector of $\mathbf{F}$ increases while the y-component remains constant, so $\dfrac{\partial P}{\partial x} > 0$ and $\dfrac{\partial Q}{\partial x} = 0$. Similarly, as y increases, the y-component of each vector increases while the x-component remains constant, so $\dfrac{\partial Q}{\partial y} > 0$ and $\dfrac{\partial P}{\partial y} = 0$.

(a) $\operatorname{div}\mathbf{F} = \dfrac{\partial P}{\partial x} + \dfrac{\partial Q}{\partial y} + \dfrac{\partial R}{\partial z} = \dfrac{\partial P}{\partial x} + \dfrac{\partial Q}{\partial y} + 0 > 0$

(b) $\operatorname{curl}\mathbf{F} = \left(\dfrac{\partial R}{\partial y} - \dfrac{\partial Q}{\partial z}\right)\mathbf{i} + \left(\dfrac{\partial P}{\partial z} - \dfrac{\partial R}{\partial x}\right)\mathbf{j} + \left(\dfrac{\partial Q}{\partial x} - \dfrac{\partial P}{\partial y}\right)\mathbf{k} = (0-0)\,\mathbf{i} + (0-0)\,\mathbf{j} + (0-0)\,\mathbf{k} = \mathbf{0}$

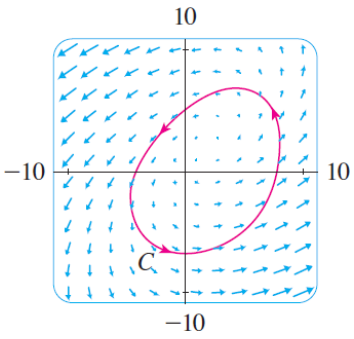
(iii) If the vector field is $\mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j} + R\,\mathbf{k}$, then we know $R = 0$. In addition, the y-component of each vector of $\mathbf{F}$ is 0, so $Q = 0$, hence $\dfrac{\partial Q}{\partial x} = \dfrac{\partial Q}{\partial y} = \dfrac{\partial Q}{\partial z} = \dfrac{\partial R}{\partial x} = \dfrac{\partial R}{\partial y} = \dfrac{\partial R}{\partial z} = 0$. P increases as y increases, so $\dfrac{\partial P}{\partial y} > 0$, but P doesn't change in the x- or z-directions, so $\dfrac{\partial P}{\partial x} = \dfrac{\partial P}{\partial z} = 0$.

(a) $\operatorname{div}\mathbf{F} = \dfrac{\partial P}{\partial x} + \dfrac{\partial Q}{\partial y} + \dfrac{\partial R}{\partial z} = 0 + 0 + 0 = 0$

(b) $\operatorname{curl}\mathbf{F} = \left(\dfrac{\partial R}{\partial y} - \dfrac{\partial Q}{\partial z}\right)\mathbf{i} + \left(\dfrac{\partial P}{\partial z} - \dfrac{\partial R}{\partial x}\right)\mathbf{j} + \left(\dfrac{\partial Q}{\partial x} - \dfrac{\partial P}{\partial y}\right)\mathbf{k} = (0-0)\,\mathbf{i} + (0-0)\,\mathbf{j} + \left(0 - \dfrac{\partial P}{\partial y}\right)\mathbf{k} = -\dfrac{\partial P}{\partial y}\mathbf{k}$

Since $\dfrac{\partial P}{\partial y} > 0$, $-\dfrac{\partial P}{\partial y}\mathbf{k}$ is a vector pointing in the negative z-direction.

2. Consider the following vector fields $\boldsymbol{F}$ in the xy-plane.



Is it possible to find a real valued function f such that $\boldsymbol{F} = \nabla(f)$?

Solution

The vectors in the Figure that start on the closed curve all appear to point in roughly the same direction as C. So it looks as if $\int_C \boldsymbol{F}(\boldsymbol{r}) \cdot d\boldsymbol{r} > 0$ and therefore it is impossible find a real value function f such that $\boldsymbol{F} = \nabla(f)$.

3. Let C be a smooth curve defined by a vector function $\boldsymbol{r}(t)$, $a \le t \le b$. Show that

(a) $\int_C \boldsymbol{v} \cdot d\boldsymbol{r} = \boldsymbol{v} \cdot \big(\boldsymbol{r}(b) - \boldsymbol{r}(a)\big)$ if v is a constant vector,

(b) $\int_C \boldsymbol{r} \cdot d\boldsymbol{r} = \frac{1}{2}\Big(|\boldsymbol{r}(b)|^2 - |\boldsymbol{r}(a)|^2\Big)$.

Solution

(a). Let $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3\rangle$. Then

$$\begin{aligned}
\int_C \mathbf{v} \cdot d\mathbf{r} &= \int_a^b \langle v_1, v_2, v_3\rangle \cdot \langle x'(t), y'(t), z'(t)\rangle \, dt = \int_a^b [v_1 x'(t) + v_2 y'(t) + v_3 z'(t)] \, dt \\
&= \big[v_1 x(t) + v_2 y(t) + v_3 z(t)\big]_a^b = [v_1 x(b) + v_2 y(b) + v_3 z(b)] - [v_1 x(a) + v_2 y(a) + v_3 z(a)] \\
&= v_1 [x(b) - x(a)] + v_2 [y(b) - y(a)] + v_3 [z(b) - z(a)] \\
&= \langle v_1, v_2, v_3\rangle \cdot \langle x(b) - x(a), y(b) - y(a), z(b) - z(a)\rangle \\
&= \langle v_1, v_2, v_3\rangle \cdot [\langle x(b), y(b), z(b)\rangle - \langle x(a), y(a), z(a)\rangle] = \mathbf{v} \cdot [\mathbf{r}(b) - \mathbf{r}(a)]
\end{aligned}$$

(b). If $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$ then

$$\begin{aligned}
\int_C \mathbf{r} \cdot d\mathbf{r} &= \int_a^b \langle x(t), y(t), z(t)\rangle \cdot \langle x'(t), y'(t), z'(t)\rangle \, dt = \int_a^b [x(t) x'(t) + y(t) y'(t) + z(t) z'(t)] \, dt \\
&= \big[\tfrac{1}{2}[x(t)]^2 + \tfrac{1}{2}[y(t)]^2 + \tfrac{1}{2}[z(t)]^2\big]_a^b \\
&= \tfrac{1}{2}\left\{\big([x(b)]^2 + [y(b)]^2 + [z(b)]^2\big) - \big([x(a)]^2 + [y(a)]^2 + [z(a)]^2\big)\right\} \\
&= \tfrac{1}{2}\left[|\mathbf{r}(b)|^2 - |\mathbf{r}(a)|^2\right]
\end{aligned}$$

4. (a) Find a function f such that $\boldsymbol{F} = \nabla(f)$. and (b) use part (a) to evaluate along the given curve .

(i). $\mathbf{F}(x, y) = xy^2\,\mathbf{i} + x^2y\,\mathbf{j}$,
C: $\mathbf{r}(t) = \left\langle t + \sin\frac{1}{2}\pi t,\ t + \cos\frac{1}{2}\pi t \right\rangle, \quad 0 \leqslant t \leqslant 1$

(ii). $\mathbf{F}(x, y) = (1 + xy)e^{xy}\,\mathbf{i} + x^2e^{xy}\,\mathbf{j}$,
C: $\mathbf{r}(t) = \cos t\,\mathbf{i} + 2\sin t\,\mathbf{j}, \quad 0 \leqslant t \leqslant \pi/2$

(iii). $\mathbf{F}(x, y, z) = yz\,\mathbf{i} + xz\,\mathbf{j} + (xy + 2z)\,\mathbf{k}$,
C is the line segment from $(1, 0, -2)$ to $(4, 6, 3)$

Solution

(i). (a) $f_x(x,y) = xy^2$ implies $f(x,y) = \frac{1}{2}x^2y^2 + g(y)$ and $f_y(x,y) = x^2y + g'(y)$. But $f_y(x,y) = x^2y$ so $g'(y) = 0 \quad \Rightarrow$ $g(y) = K$, a constant. We can take $K = 0$, so $f(x,y) = \frac{1}{2}x^2y^2$.

(b) The initial point of C is $\mathbf{r}(0) = (0,1)$ and the terminal point is $\mathbf{r}(1) = (2,1)$, so

$\int_C \mathbf{F} \cdot d\mathbf{r} = f(2,1) - f(0,1) = 2 - 0 = 2.$

(ii). (a) $f_y(x,y) = x^2e^{xy}$ implies $f(x,y) = xe^{xy} + g(x) \quad \Rightarrow \quad f_x(x,y) = xye^{xy} + e^{xy} + g'(x) = (1+xy)e^{xy} + g'(x)$. But $f_x(x,y) = (1+xy)e^{xy}$ so $g'(x) = 0 \quad \Rightarrow \quad g(x) = K$. We can take $K = 0$, so $f(x,y) = xe^{xy}$.

(b) The initial point of C is $\mathbf{r}(0) = (1,0)$ and the terminal point is $\mathbf{r}(\pi/2) = (0,2)$, so

$\int_C \mathbf{F} \cdot d\mathbf{r} = f(0,2) - f(1,0) = 0 - e^0 = -1.$

(iii). (a) $f_x(x,y,z) = yz$ implies $f(x,y,z) = xyz + g(y,z)$ and so $f_y(x,y,z) = xz + g_y(y,z)$. But $f_y(x,y,z) = xz$ so $g_y(y,z) = 0 \quad \Rightarrow \quad g(y,z) = h(z)$. Thus $f(x,y,z) = xyz + h(z)$ and $f_z(x,y,z) = xy + h'(z)$. But $f_z(x,y,z) = xy + 2z$, so $h'(z) = 2z \quad \Rightarrow \quad h(z) = z^2 + K$. Hence $f(x,y,z) = xyz + z^2$ (taking $K = 0$).

(b) $\int_C \mathbf{F} \cdot d\mathbf{r} = f(4,6,3) - f(1,0,-2) = 81 - 4 = 77.$